Generalized linear models

Basic concepts of regression

Prof. Dr. Andreas Groll

Summer term 2026

Basic concepts of regression - Contents

Components of regression

Encoding of the independent variables

Classical linear regression model

Components of regression

Fundamental problem

How do variables $x_1, \ldots, x_p$ affect a dependent variable y?

$$x_1, \ldots x_p \qquad \rightarrow \qquad y$$

Components of regression

Questions:

1. What is the difference to correlation?

2. What is the difference to statistical testing?

What are the main objectives?

- Detection of interdependency
- Quantification of the relation
- Prediction / prognosis

What is being investigated?

The variable $y \mid \mathbf{x}$ with $\mathbf{x} = (x_1, \ldots, x_p)^\top$ is examined.

Particularly:

- $f_{y|\mathbf{x}}$ or $f_y(\mathbf{x})$
- $F_{y|\mathbf{x}}$ or $F_y(\mathbf{x})$
- $\mathbb{E}_{y|\mathbf{x}}$or $\mathbb{E}(y \mid \mathbf{x})$
- $Q_T(y \mid \mathbf{x})$

Structures

Structure of the dependent variable

- y metrical

Structures

- Special case: y skewed distribution

Structures

- y binary

Structures

- y multiple-categorical

Structures

Structure of the explanatory term

- Simplest option: linear

- More general: additive models

Structures

- Additive model with interactions

- Additive model with an effect-modified variable

Encoding of the independent variables

Linear model

$$\mathbb{E}(y \mid \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

1. x_j metrical, e.g.:
   - y :
   - x_1 :
   - β_1 :

For multiple independent variables

- One independent variable, e.g.
  - y :
  - x_1 :
  - β_1 :
- Two independent variables, e.g.
  - y :
  - x_1 :
  - x_2 :
  - β_1 :
  - β_2 :

For multiple independent variables

- β_j is interpretable given adhered other variables (adjustment)

- marginal correlation $\neq$ conditional correlation

Encoding of the independent variables

2. x_j binary
   - y :
   - x_1 :
   - $\beta_1 =$

Encoding of the independent variables

3. x_j multiple-categorical, e.g.
   - y :
   - x_1 :

Dummy coding: 0/1-Encoding

- $x_j \in 1, \ldots k$
- $x_{j(i)} =$

Dummy coding: 0/1-Encoding

- Model: $\mathbb{E}(y \mid x) = \beta_0 + \beta_{j(1)} x_{j(1)} + \cdots + \beta_{j(k)} x_{j(k)}$
- **Problem:**

Dummy coding: Identifiability

- To achieve uniqueness of the parameters, restrictions are required
- E.g., one category is set to 0
- Convention:
  - Set category k to 0 $\rightarrow \beta_{j(k)} = 0$
    $\rightarrow$ reference category

Dummy coding: Identifiability

Example: $k = 3$

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_{1(1)} x_{1(1)} + \beta_{1(2)} x_{1(2)} + \beta_{1(3)} x_{1(3)}$$

- $\mathbb{E}(y \mid x_1 = 1) =$
- $\mathbb{E}(y \mid x_1 = 2) =$
- $\mathbb{E}(y \mid x_1 = 3) =$

Effect encoding

- $x_j \in 1, \ldots k$
- Model: $\mathbb{E}(y \mid x) = \beta_0 + \beta_{j(1)} x_{j(1)} + \cdots + \beta_{j(k)} x_{j(k)}$

- **Problem:**

Effect encoding: Identifiability

- Restriction: $\sum_{i=1}^{k} = \beta_{j(i)} = 0$
- One has:

$\rightarrow$ β_0 is

Effect encoding

Notation:

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_{1(1)} x_{1(1)} + \cdots + \beta_{1(k-1)} x_{1(k-1)}$$

Then:

- $\mathbb{E}(y \mid x_1 = 1)$
- $\mathbb{E}(y \mid x_1 = k - 1)$
- $\mathbb{E}(y \mid x_1 = k)$

Dummy and effect encoding

Which encoding is useful in which situation?

- Dummy coding

- Effect encoding

Linear predictor

$$\eta = \mathbf{x}^\top \boldsymbol{\beta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Variables can

- be metrical, encoded or both of it
- include polynomial terms etc., e.g. age^2

Predictor characterization via operators/ Wilkinson-Rogers notation

Let $x, \tilde{x}$ metrical, j categorial

Operator	Abbreviation	Linear term
	x	$x\beta$
	j	$x_{j(1)}\beta_{j(1)} + \cdots + x_{j(k-1)}\beta_{j(k-1)}$
$+$		
$\cdot$		
$*$		

Classical linear regression model

Linear regression model

$$y_i = \mathbf{x}_i^\top \beta + \varepsilon_i$$

with

1. $\mathbb{E}(\varepsilon_i) = 0$
2. $\text{Var}(\varepsilon_i) = \sigma^2$
3. $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$
4. $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$

Classical linear regression model - alternative notation

- Random component:

$$y_i \mid \mathbf{x}_i \sim \mathcal{N}(\mu_i, \sigma^2)$$

- Systematical component:

$$\mu_i = \mathbb{E}(y_i \mid \mathbf{x}_i) = \mathbf{x}_i^\top \beta$$